

vapor

A certified tensor compiler and the laboratory built on it

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Architecture and guarantees

The principle

Every result carries what is needed to judge it.

- a kernel: evidence that it computes the oracle's bits, on every substrate
- an optimal integer program: the branch-and-bound tree, checked by separate code
- a refuted conservation law: the polynomial remainder and the point where it is not zero
- a statistical claim: a control that a broken implementation would have passed

Three properties: **determinism**, **certification**, **containment**.

Layers

Interfaces: CLI (Dīwān) · console · TUI · MCP · LSP · Al-Qalam

Laboratory: certified deciders · Almizan · Alembic/Athamor · Crucible · Assay · workbench · finance

model airlock

substrate airlock

network airlock

Engine: sessions, batching, paged cache, recurrence, speculation, receipts

Core: algebra $\rightarrow$ rewrite $\rightarrow$ KIR $\rightarrow$ selection $\rightarrow$ allocation (Lean-checked) $\rightarrow$ encoders $\rightarrow$ ladder $\rightarrow$ certificate

worker: x86-64, AArch64, RV64GCV

fabric: Vulkan

oracle: exact binary32

101k lines of Elixir (core: 10k) · 4.7k Zig · 1.5k Lean (82 theorems) · no dependencies outside OTP · no NIF

Programs are terms; meaning is the oracle's

- a symbolic free algebra: element-wise ops, sel, contractions (linear, grouped, masked, 4-bit, int8), reductions, gather_row, rope, kv_write, attention, sample
- semi-dynamic dimensions: bounds at the maximum, extended by monotonicity lemmas (Lean)
- recurrent programs: the loop runs in the worker; the state never leaves it
- Vapor.F32: exact binary32 on the BEAM (fma in dyadic rationals, one rounding)

:canonical	separate rounding; one 16-lane reduction tree; $\div$ correctly rounded $\Rightarrow$ identical bits on every substrate
:fast	fused FMA $\Rightarrow$ every output inside a rigorous envelope (exact dyadic bounds)

Parallelism from **independent outputs**, never from reassociating a sum.

The compiler



- rewriting admits only identities valid over all of IEEE-754 ($x + (+0) \not\rightarrow x$)
- one kernel IR over typed virtual registers; g generalises LMUL to AVX2, AVX-512, NEON, RVV
- out of registers? smaller g , fewer rows, or cut the fused region: **never spill**
- every allocation revalidated by `check_alloc`, extracted from Lean with `checkAlloc_sound`
- encoders write bytes (REX/VEX/EVEX, A64, RV64GCV, SPIR-V); `binutils` and `spirv-val` only check

Substrates and isolation

vapor-worker (static Zig, no libc, 350 KB)

- seccomp allow-list, NO_NEW_PRIVS, W^X pages, watchdog
- thread pool and counters set up before the filter
- native code, or the RVV 1.0 interpreter (VLEN 128–512, poison mode)
- resident sessions: weights mapped once, state kept

vapor-fabric: Vulkan compute, its own process

One protocol: {packet, 4} frames on stdio

- HELLO · RUN · EMIT · DONE · ERR · OPEN/STEP/CLOSE
- a crash kills only the process; the next unit runs

Memory: weights in /dev/shm/vapor-<sha256>, mapped copy-on-write; imported without copy by Vulkan **Failover:** fabric → AVX-512 → base → RVV interpreter → oracle: same bits at every link

Parallelism without changing bits

form	where	invariant
SIMD	AVX2, AVX-512, NEON, RVV, SPIR-V	= oracle
rows per iteration	chosen by the allocator	= oracle
threads	the worker's pool	$1 = n$ threads
continuous batching	decode + prefill chunks in one step	alone = in a batch
paged cache	block tables	paged = contiguous
replicas	one compilation, shared pages	any replica
speculation	$k+1$ rows verified in one step	= target alone

Speculative decoding needs no extra proof: the verify step is bit-identical to single steps.

The verification ladder

1	sorts, shapes	<code>Program.check/1</code>
2	admission	checker accepts the allocation on <i>every</i> ISA; no int8 wrap at the max extents
3	adjoint	$\langle Wx, v \rangle = \langle x, W^T v \rangle$ catches layout errors
4	oracle	each CPU substrate = exact oracle, bit for bit
5	parity, envelope	equal digests across substrates; outputs inside the envelope
6	certificate	Ed25519; host-independent payload $\Rightarrow$ byte-identical across nodes $\Rightarrow$ co-signable, quorum

Lean 4 (core only, no sorry/axiom): integer parity of any reduction tree, bank injectivity, the affine monoid, allocation soundness, envelope, Higham's bound $\rightarrow$ extracted at build time, 480+ conformance vectors.

Models: function versus spelling

tier	names
core	none (atom-table test)
topologies	what they compute: decoder, delta-rule hybrid, encoder, encoder-decoder, Mamba, DiT, U-Net, VAE
spellings	families, as data : aliases and refusals
agent protocols	the vendor's wire format

- one owner per checkpoint; no owner $\Rightarrow$ refusal listing every near miss and its fix
- the program checked against its contract at the boundary
- alias < blueprint < topology, in cost
- **preflight**: verdict from config.json and safetensors headers alone

A frontier topology: the delta-rule hybrid

$$S_t = (I - \beta_t k_t k_t^\top) \text{Diag}(\alpha_t) S_{t-1} + \beta_t k_t v_t^\top, \quad \alpha_t = e^{g_t}, \quad g_t = g_{\min} \sigma(e^A z_t)$$

- chunked prefill via the UT transform $T = (I + \text{diag}(\beta)L_s)^{-1}$: equal to the recurrence to $3 \cdot 10^{-16}$
- latent attention: only the latent cached; key map absorbed into the query
- latent mixture of experts; soft-capped gated activation; **MXFP4 decodes exactly** to binary32
- vs an independent float64 reference: rel. error $4.4 \cdot 10^{-6}$; greedy identical; native = oracle; controls fail
- balanced routing is a totally unimodular b -matching: **certified** by the exact simplex

The network airlock

Rule: an agent never connects on its own; only the person opens the tap.

- fetchers the person declares (`siphons.json`): any program
- a port with a reduced environment, deadline, byte cap
- through the format airlocks; SHA-256 pins; a receipt
- agents: `siphon_propose` queues, fetches nothing

Streaming weights: time per token $\geq$ active bytes / bandwidth

	15 GB dense	146 GB active
RAM 50 GB/s	0.3 s	3 s
NVMe 3 GB/s	5 s	50 s
net 20 MB/s	12.5 min	2 h

$\Rightarrow$ read-once work (prefill), not conversation

Deciders: a search proposes, a checker decides

SAT	CDCL; model, or DRUP refutation checked by separate code
LP / MIP	exact simplex over $\mathbb{Q}$: duals, Farkas vectors; branch-and-bound tree as certificate
Gröbner, Knuth–Bendix	basis $\{1\}$ or remainder; convergent rewriting with derivations
causes	ID algorithm: estimand, or a hedge; on a <i>stated</i> diagram
positivity	Bernstein subdivision in exact integers, or the refuting point
circuits, netlists	truth table or SAT miter with DRUP; sky130 mapping proved before printing
contracts	antinomies with a scenario; consistency by DRUP

A language model's proposal goes through the same checker as anyone's.

Almizan: claims that do not compile until decided

```
(claim energy (root H-f-Z) (wzn burhan)
  (inputs (x q) (v q))
  (field (x v) (v (- x))))
(proof conserved)
(body (+ (kinetic v) (* 1/2 x x))))
```

(دعوى طاقة (جذر ح-ف-ظ) (وزن برهان)
(برهان محفوظ)

- root (kind) + pattern (use): a morphological type system
- Latin and Arabic: two printings, one tree, one hash
- conserved: $\dot{H} = \nabla H \cdot f$ over $\mathbb{Q}$;
damped $\Rightarrow \dot{H} = -\frac{1}{10}v^2$ at $v=1$
- identities, positivity, invariants (SAT), causal queries
- export to core Lean, checked to agree

Containment of open input

Alembic: problems as text

- fuel on every call and iteration; bounded depth
- integers $\leq 65,536$ bits; lists ≤ 2 million
- no atoms created; no I/O, clock or randomness
- a literal reader for data

Athantor: portfolio search with a random control; journal under a Merkle root; the

Touchstone re-checks every certificate

The hermetic seal

- one primitive for every job on untrusted input
- heap cap that *counts off-heap binaries*
- deadline; the caller survives a memory bomb

Agents: effect classes (pure, observe, act), grants in the digest, write-ahead journal, replay is verification

The evidence discipline

Every check = **value** + **control** + **threshold**: *would this answer come out with nothing to find?*

check	value	control
gradient furnace	$a(\frac{1}{2} + n_y/3)$	biased estimator: caught
ODE	Dormand–Prince 10^{-8}	RK4 $h=0.1$: $> 10^{-5}$
SAT	witness + DRUP	truncated proof: rejected
scaling law	planted exponent in CI	shuffled losses: fails
ink rendering	shadow = 0.8×0.3	no occluder: lit

Assurance ledger (proved / checked / tested / owed) generated into the docs; docs that name a missing module or path fail a test; test tiers excluded only when their tool is absent; a content-addressed cache reruns only what a change can reach.

Interfaces

```
vapor wzn check energy.wzn  
vapor athanor run p.nbq | vapor verify p.nbq -  
vapor logic knapsack.lp | jq .objective  
vapor qalib check spec.net synth.v  
vapor render scene.txt --ink  
vapor siphon preflight hub config.json ...  
vapor qalam energy.wzn
```

exit 0 positive · 1 negative · 2 usage · 3 bad input ·
4 failure; JSON in pipes

- one interpreter for CLI, TUI and the console's terminal (jailed)
- console: one offline page, live furnace, GPU render desk
- MCP: 27 tools; agents propose, checkers decide
- LSP: verdicts as diagnostics and inlay hints
- **Al-Qalam**: vi keys, verdicts in the gutter, scrubable numbers, Merkle undo tree

Limits

- RVV and NEON validated under QEMU, Vulkan under lavapipe; no physical RVV, discrete GPU or Apple Silicon run
- transcendental functions identical everywhere but within 1–2 ulps, not correctly rounded
- 4-bit GEMV bound by instruction issue; the Vulkan path is correct, not tuned
- frontier topologies checked on small instances; released spellings read through header-only admission
- hosted models are observations, not functions
- a certificate says what was computed, not whether a model is good; a causal verdict needs its diagram
- the seal bounds memory and time, not code with access to the BEAM

vapor

Determinism · certification · containment

`docs/ARCHITECTURE.md · docs/ASSURANCE.md · technical/monograph.pdf`